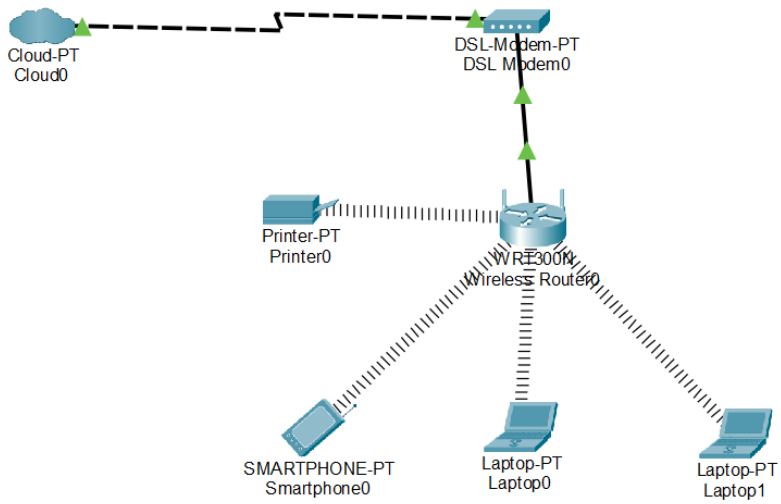
# BUILD SOHO NETWORK

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Ans: -




Connection Details:

1. Broadband Internet connects to the Wi-Fi router using a wired connection.
2. Laptop 0 connects to the router using Wi-Fi (Wireless).
3. Laptop 1 connects to the router using Wi-Fi (Wireless).
4. Smart Phone connects to the router using Wi-Fi (Wireless).
5. Printer connects to the router using an Ethernet cable (Wired).

Wireless (Wi-Fi) Connections:

1. Wireless Router → Printer0
2. Wireless Router → Smartphone0
3. Wireless Router → Laptop0
4. Wireless Router → Laptop1

2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.<br><br>Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Ans: -

| No. | Hardware Component | Brand / Model Suggestion | Qty. | Approx. Price (₹) |
|---|---|---|---|---|
| **1** | Broadband Modem / ONT | JioFiber / Airtel ONT | 1 | 1,500–3,000 |
| **2** | Wi-Fi Router | TP-Link Archer AX23 | 1 | 4,000–5,500 |
| **3** | Laptop | Dell / HP Laptop | 1 | 35,000–45,000 |
| **4** | Smart TV | Samsung / LG Smart TV (32-inch) | 1 | 12,000–18,000 |
| **5** | Ethernet Cable | CAT6 D-Link / FEDUS | 2 | 200–500 |
| **6** | RJ45 Connectors | D-Link / FEDUS | 4 | 100–300 |
| **7** | Wi-Fi Range Extender | TP-Link RE315 | 1 | 2,500–3,500 |
| **8** | Network Switch | TP-Link TL-SG1005D (5-Port) | 1 | 1,200–1,500 |
| **9** | UPS / Power Backup | APC Back-UPS | 1 | 4,000–7,000 |

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Ans: -

## Step-by-Step Instructions to Configure Wi-Fi Router for SOHO Network

* Router Model: TP-Link Archer AX23
- Connect the Router: Connect the broadband modem/ONT to the Wi-Fi router using an Ethernet cable. Switch on both devices.
- Connect to Router: Connect your laptop to the router using Wi-Fi or an Ethernet cable.
- Open Router Settings: Open a web browser and enter 192.168.0.1 or tplinkwifi.net in the address bar.
- Login to Admin Panel: Enter the default login details or create an admin password if requested.
- Set a Unique SSID: Go to Wireless Settings and enter a unique Wi-Fi network name (SSID), such as Bhargav_SOHO_WiFi.
- Enable WPA2 Security: Select WPA2-PSK (AES) as the security option and create a strong Wi-Fi password.
- Change Admin Password: Open System Tools / Administration and change the default admin password to a strong, unique password.
- Save Settings: Click Save or Apply to save all the changes. The router may restart automatically.
- Reconnect Devices: Connect your laptop and Smart TV to the new Wi-Fi network using the updated password.
- Test Internet Connection: Open a website on your laptop and Smart TV to check whether the internet is working properly.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Ans: -

## ∗ IP Address Configuration for SOHO Network

The SOHO network uses static IP addresses for the printer and Smart TV, while laptops receive IP addresses automatically through DHCP.

➢ Network Configuration:

- Router IP Address: `192.168.1.1`
- Subnet Mask: `255.255.255.0`
- DHCP Range: `192.168.1.100 – 192.168.1.199`

### ➢ IP Address Assignment Table

| No. | Device | MAC Address (Example) | IP Address | Configuration |
|---|---|---|---|---|
| 1 | Wi-Fi Router | 02:11:22:33:44:01 | 192.168.1.1 | Static |
| 2 | Printer | 02:11:22:33:44:02 | 192.168.1.10 | Static |
| 3 | Smart TV | 02:11:22:33:44:03 | 192.168.1.20 | Static |
| 4 | Laptop | 02:11:22:33:44:04 | 192.168.1.100 | DHCP |

### ➢ Steps to Configure IP Addresses

1. Configure the Router

- Open the router settings using `192.168.1.1`.
- Log in with the admin password.
- Open DHCP Settings.
- Enable the DHCP server.
- Set the DHCP range to `192.168.1.100 – 192.168.1.199`.

2. Assign Static IP to Printer

- Open the printer's Network Settings.
- Select Manual IP Configuration.
- Enter IP address: `192.168.1.10`.
- Enter Subnet Mask: `255.255.255.0`.
- Enter Gateway: `192.168.1.1`.

3. Assign Static IP to Smart TV

- Open Smart TV Network Settings.
- Select Manual IP Settings.
- Enter IP address: `192.168.1.20`.
- Enter Subnet Mask: `255.255.255.0`.
- Enter Gateway: `192.168.1.1`.

4. Configure Laptop Using DHCP

- Open Network Settings on the laptop.
- Select IP Assignment: Automatic (DHCP).
- The router will automatically assign an IP address to the laptop.

5. Test the Network

- Open Command Prompt on the laptop.
- Type `ipconfig` to check the assigned IP address.
- Type `ping 192.168.1.10` to test the printer.
- Type `ping 192.168.1.20` to test the Smart TV.

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Ans: -

## * Troubleshooting Laptop Internet Connection in SOHO Network

### Possible Causes and Solutions

#### * Wi-Fi Connection Problem

- Cause: The laptop may be disconnected from Wi-Fi or connected to the wrong network.
- Diagnosis: Check the Wi-Fi icon and verify the SSID.
- Solution: Connect to the correct Wi-Fi network (`Bhargav_SOHO_WiFi`) using the correct password.

#### * Incorrect IP Address

- Cause: The laptop may not receive a valid IP address from the router through DHCP.
- Diagnosis: Open Command Prompt and type `ipconfig` to check the IP address.
- Solution: Run `ipconfig /release` and `ipconfig /renew` to obtain a new IP address automatically.

#### * Router Problem

- Cause: The Wi-Fi router may be frozen or not working properly.
- Diagnosis: Check the router's power and internet indicator lights.
- Solution: Restart the router and reconnect the laptop to Wi-Fi.

## * DNS Server Problem

- Cause: Incorrect DNS settings may prevent websites from opening.
- Diagnosis: Open Command Prompt and type `ping 8.8.8.8.` If this works but websites do not open, DNS may be the problem.
- Solution: Set the DNS server to `8.8.8.8` (Google DNS) and try opening the website again.

## * Internet Service Provider (ISP) Problem

- Cause: The broadband internet connection may be down.
- Diagnosis: Check whether other devices, such as the Smart TV, can access the internet.
- Solution: Restart the modem/ONT and contact the ISP if the internet is still unavailable.

## * Outdated or Faulty Wi-Fi Driver

- Cause: The laptop's Wi-Fi network driver may be outdated or corrupted.
- Diagnosis: Open Device Manager and check Network Adapters for any warning symbols.
- Solution: Update or reinstall the Wi-Fi driver and restart the laptop.

## * Firewall Blocking Internet Access

- Cause: Windows Firewall or antivirus software may block network traffic.
- Diagnosis: Check Windows Firewall settings and antivirus network protection.
- Solution: Allow the required network connection through the firewall and check internet access again.